

Khwaja Yunus Ali University

School of Science and Engineering

Department of Computer Science and Engineering

Final Examination: Spring Semester 2020

Program: B.Sc. in CSE (Batch: 10th) 1st Year 1st Semester

Course Title: Physics – I

Course Code: PHY 1101

Total Time: 90 min

Full Marks: 40

Part-A

[Write the no. of the appropriate answer of the questions sequentially] 10 × 1 = 10

- What is the unit of the reduced mass?
a) $C^2 N^{-1} m^{-2}$ b) $N m^2 C^{-2}$
c) $T m A^{-1}$ d) C
e) kg
- 1 Tesla = ? Gauss
a) 10 b) 100
c) 1000 d) 10000
e) 100000
- A force of 4N on the spring produces a displacement of 0.02m. What is the force constant of the spring?
a) $400 N m^{-1}$ b) $200 N m^{-1}$
c) $100 N m^{-1}$ d) $0.08 N m^{-1}$
e) $0.05 N m^{-1}$
- The direction of induced current will be anticlockwise, when a magnet the coil.
a) N-pole move away from b) S-pole move nearer to
c) N-pole move nearer to d) N-pole fixed with
e) S-pole fixed with
- If the closed path does not link with the current loop, then the Ampere's law –
a) $\oint_l \vec{B} \cdot d\vec{l} = 0$ b) $\oint_l \vec{B} \cdot d\vec{l} = \mu_0$
c) $\oint_l \vec{B} \cdot d\vec{l} = i$ d) $\oint_l \vec{B} \cdot d\vec{l} = n$
e) $\oint_l \vec{B} \cdot d\vec{l} = BiL$
- The maximum acceleration of SHM is
a) $-\omega x$ b) $-\omega A$
c) $-\omega^2 A$ d) $-\omega^2$
e) $-A$
- What is the value of form factor of AC?
a) 0.01 b) 0.10
c) 0.11 d) 1.11
e) 11.1
- Which one is right for the power of the primary (P_p) and secondary (P_s) coil of transformer?
a) $P_p > P_s$ b) $P_p < P_s$
c) $P_p \gg P_s$ d) $P_p \ll P_s$
e) $P_p = P_s$
- How long will take to reach the peak from zero of AC signal which is supplied in our home electric line in Bangladesh?
a) $0.005 sec$ b) $0.02 sec$
c) $25 sec$ d) $50 sec$
e) $220 sec$
- Average value of alternating current for full cycle is
a) 7 b) 0.707
c) 0.673 d) 0.637
e) Zero

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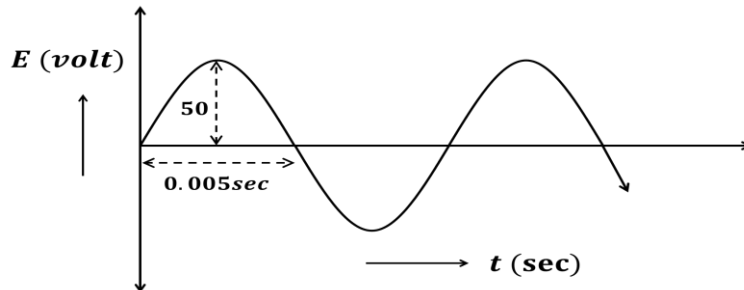
Full Marks: 40

Part-B

[Answer any five (05) questions out six (06)]

5 × 6 = 30

1. See diagram of AC current given below and answer the following question:



- a) Calculate the average, rms value & time period of above-mentioned signal. 3
- b) Is it possible to drive an equation for above curve? – Justify your answer. 3
2. a) Formulate the Newman's Law. 2
- b) Write down the basic differences between the step-up and step-down transformer. 2
- c) Average emf induced in a coil is 32V when current decreases from 10A to 2A in 0.1sec in the coil. Calculate the self – inductance in the coil. 2
3. a) Prove that, $x = A \sin(\omega t + \delta)$ is the solution of differential equation of SHM. 3
- b) Find the time period of oscillation having 0.01m amplitude and 0.8ms^{-1} maximum velocity. 3
4. a) Draw the kinetic & potential energy curve with respect to displacement for the object in SHM. 2
- b) Write a short note on: **(Any two)** 4
- i) Damped oscillation
- ii) Reduce Mass
- iii) Resonance
- iv) Lissajous Figures
5. a) Define magnetic moment and write down the relation of it with torque. 2
- b) A solenoid is 1.0m long and 3.0cm in mean diameter. It has five layers of windings of 850 turns each and carries current of 5.0amp. (a) What is B at its centre? (b) What is the magnetic flux ϕ_B for a cross-section of the solenoid at this centre? 4
6. a) Draw the representation of the direction of magnetic field in the plane of paper. 2
- b) Find the force of each segment of the wire: 3

Conducting wire in a magnetic field

